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Therapeutic covering for horses

Abstract

A therapeutic covering for a horse having a mesh material that is infused with ceramic mineral powder (emitting far-infrared wavelengths) and magnets coupled to the mesh material. The mesh material can have a mesh size that provides therapeutic effect, is breathable, and is bug resistant.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of U.S. Non-provisional patent application Ser. No. 16/897,948, filed Jun. 10, 2020, which is a continuation of U.S. Non-provisional patent application Ser. No. 15/221,244 filed on Jul. 27, 2016, which claims the benefit of U.S. Provisional Patent Application No. 62/211,305 filed Aug. 28, 2015, the contents of which applications are herein incorporated by reference in their entirety.

FIELD OF THE DISCLOSURE
(2) The present disclosure generally relates to the field of magnetic and ceramic textile therapies, and more specifically relates to a therapeutic covering for equine use.
BACKGROUND

(3) Competition horses are tested for illegal substances before competitions. In some competitions, there is a 12-hour administration rule for any medication prior to competition. Training can cause soreness in a horse, and the horse may need help with discomfort associated with training and even injuries resulting from competition and training. For example, for back soreness of a horse, the most commonly prescribed medication for soreness relief is methocarbamol, which is a muscle relaxant. The maximum dosage of methocarbamol currently allowed is 5.0 g/1000 lbs, and methocarbamol is administered orally or intravenously at least 12 hours prior to competition. There are also restrictions on NSAIDS and natural anti-inflammatory drugs prior to competition for horses.
(4) There is a need for an effective, non-invasive, non-medicinal, and non-detectable technique for treating horse soreness and discomfort.

SUMMARY

(5) Disclosed is a therapeutic covering for a horse comprising a mesh material infused with a ceramic mineral powder, and a plurality of magnets coupled to an outer surface of the mesh material, wherein the mesh material has a mesh size in a range of greater than 2.5 millimeters and less than about 5 millimeters.
(6) Also disclosed herein is a method for ceramic and magnetic therapy on a horse. The method can include placing the therapeutic covering on a horse; and allowing the therapeutic covering to perform ceramic and magnetic therapy on the horse. The therapeutic covering can be of any embodiment disclosed herein. The method can further include positioning the therapeutic covering on the horse such that the magnets are positioned to correspond with acupuncture points of the horse.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The figures which accompany the written portion of this specification illustrate embodiments and method(s) of use for the therapeutic covering:
(2) FIG. **1** is a side view illustrating a therapeutic covering on a horse according to the disclosure.
(3) FIG. **2** is a perspective view of the therapeutic covering and horse of FIG. **1**.
(4) FIG. **3** is an isolated plan view illustrating a pocket of the therapeutic covering having a magnet therein.
(5) The various embodiments of the disclosure will hereinafter be described in conjunction with the appended drawings.

DETAILED DESCRIPTION

(6) Disclosed herein is a therapeutic covering for horses that is made of a mesh material infused with ceramic mineral powder. The therapeutic covering also has magnets that are coupled to the mesh material in positions that correspond with the primary acupuncture points of the horse or pony when the therapeutic covering is placed over the horse or pony. The therapeutic covering provides a combination therapeutic effect, is breathable, and is unexpectedly bug resistant.
(7) The ceramic particles in the ceramic mineral powder provide ceramic therapy. The ceramic particles emit far-infrared wavelengths which stimulate vibration of the oxygen atoms within the water molecules (H.sub.2O). As the oxygen atom in each water molecule vibrates around the hydrogen atoms, the water molecules begin to shrink in size which then stimulates the increase in blood circulation as well as dilation of blood vessels. Ceramic therapy can also help develop new capillaries (branching blood vessels with a hair-like thickness) for treatment of deep muscle lesions or soft tissue injuries.
(8) The magnets provide magnetic therapy. Magnetic therapy is an accepted alternative medicine practice that involves the use of static or pulsing magnetic fields to derive beneficial health effects

such as oxygenating soft tissues, stimulating blood flow in underlying tissues, improving overall immunity, and creating the harmonization of bodily functions.
(9) The combination of ceramic therapy and magnetic therapy provided by the disclosed therapeutic covering offers a unique, safe, and non-invasive treatment for horses. The therapeutic benefits of this combination therapy can be used as pre-exercise and/or post-exercise therapy for horses. For example, the therapeutic covering can help to “warm-up” or stimulate a horse's muscles and relax tension before exercise. Additionally or alternatively, the therapeutic covering can help to reduce soreness, reduce inflammation, and increase blood circulation for enhanced healing and recovery as a post-exercise therapy.
(10) Referring now to FIGS. **1-3**, the therapeutic covering **100** is shown placed on a horse. The therapeutic covering **100** has a mesh material **110** configured to cover at least a portion of a horse's body. The mesh material **110** is infused with ceramic mineral powder, and a plurality of magnets **120** are attached to an outer surface **111** of said mesh material **110** at predetermined locations. In embodiments, the predetermined locations can correspond to acupuncture points of the horse when the therapeutic covering **100** is placed on the horse.
(11) The mesh material **110** can be made of moisture wicking fibers that can include, but are not limited to, polyester, polypropylene, nylon, micro modal, bamboo, wool, or combinations thereof. The moisture wicking and fibers can pull moisture away from the horse's skin to help keep the horse cooler, to help the horse dry faster after being wet, or both. Moreover, the moisture wicking fibers can be chosen for the suitability to infuse ceramic powder into the fibers.
(12) In embodiments, the mesh material **110** can be a net of knitted strands of fibers. For example, the mesh material **110** can have a warp-knitted structure. In embodiments, the warp-knitted structure can be a 20-needle warp-knitted structure.
(13) In embodiments, the mesh material **110** can be a heavyweight mesh or heavyweight net. “Heavyweight” is defined as weighing more than 50 g/m (45.87 g/yd). A heavyweight mesh or net can be used to provide a light covering for the horse that can fit under or over other blankets or equipment while also being heavy enough to follow the contour of the horse while the therapeutic covering **100** is in use. In embodiments, the mesh material **110** can weigh 295 g/m (270 g/yd). The mesh material **110** can also be durable such that the mesh material **110** is hard to tear with normal usage. The mesh material **110** can have a high tensile strength allowing the covering to be pulled and grabbed during use without tearing or breaking. The mesh material **110** can have a tensile strength in the warp direction in a range of about 9.84 kg/cm (25 kg/in) to about 13.77 kg/cm (35 kg/in); alternatively, about 10.62 (27 kg/in) to about 12.6 kg/cm (32 kg/in); alternatively, a tensile strength of about 11.41 kg/cm (29 kg/in) in the warp direction. The mesh material **110** can have a tensile strength in the weft direction in a range of about 7.87 kg/cm (20 kg/in) to about 11.81 kg/cm (30 kg/in); alternatively, about 9.06 kg/cm (23 kg/in) to about 10.62 kg/cm (27 kg/in); alternatively, a tensile strength of about 9.84 kg/cm (25 kg/in) in the weft direction.
(14) The mesh material **110** is characterized by having openings (e.g., see opening **140** in FIG. **3**) between fibers or strands of fibers. The openings can have any shape, such as but not limited to circle shape, oval shape, triangle shape, square shape, rectangle shape, diamond shape, or combinations thereof. The mesh size of the mesh material **110** is defined by the size of the openings (e.g., see opening **140** in FIG. **3**) in the mesh material **110**. The openings in the mesh material **110** is large enough to be breathable. “Breathable” as used herein means the openings in the mesh material are of a size that allows horse body heat to transfer through the openings of the therapeutic covering and for ambient air to transfer through the same openings. It was found that a mesh size of 2.5 millimeters or less (e.g., the mesh size of the mesh material of a typical scrim) was too small to be breathable for a horse in a warm or hot climate that does not need to keep warm. That is, a mesh size of 2.5 millimeters or smaller led to sweating of horses when not desired. In the therapeutic covering **100** disclosed herein, the mesh size (e.g., the largest distance of opening **140**) of the mesh material **110** can be greater than 2.5 millimeters to about 5 millimeters; alternatively, greater than

2.5 millimeters to about 4 millimeters; alternatively, from about 3 millimeters to about 4 millimeters; alternatively, a width of about 3.5 millimeters; alternatively, a width of 3.5 millimeters. The “largest distance” of an opening depends on the shape of the opening. For example, the largest distance of a circle shaped opening is the diameter of the circle, the largest distance of an oval-shaped opening is the major diameter, the largest distance of a triangle-shaped opening is the longest side of the triangle, the longest distance of a square-shaped opening is any side of the square, the largest distance of a rectangular shaped opening is the longest side, the largest distance of a diamond shaped opening is the major axis.

(15) It was surprisingly found that a mesh size in the disclosed range was not only breathable, the mesh size visually reduced horse irritation from bugs (e.g., flies). The bugs unexpectedly leave the horse to pursue other horses not wearing the therapeutic covering **100**. While not being limited by theory, it is believed that the flies are deterred from targeting the horse even though the mesh size is large enough for the flies to touch the horse's fur and/or skin. While small (2.5 millimeters or less) mesh sizes can physically prevent insects and bugs from landing on the skin of the horse because the openings are smaller than insect size in some cases, it was unexpected and surprising that a mesh size of greater than 2.5 millimeters to about 5 millimeters led to the insects leaving the horse alone rather than continuing to try and land on the horse in the openings **140** of the mesh material **110**.

(16) The mesh material **110** can also be infused with a ceramic mineral powder. More particularly, the fibers of the mesh material **110** can be infused with the ceramic mineral powder. The ceramic mineral powder may be formed as a combination of lead-free materials including, but not being limited to, silica, magnesia, alumina, copper, titania, zirconia, silicon carbide, mullite, rare earth phosphates, or combinations thereof. The ceramic particles that make up the powder can be infused via any technique known in the art with the aid of this disclosure. For example, the ceramic particles can be infused by incorporating the ceramic particles into the polymers strands that make up the fabric. Additionally or alternatively, the ceramic particles can be melted into a slurry that is then contacted with the fibers of the mesh material so that the ceramic particles absorb into the fibers. Additionally or alternatively, the ceramic particles can be added to a solve to make a solution, the fabric is then placed in the solution, and then current is applied to the solution having the fibers placed therein to bond the ceramic particles to the fibers. Different methods can be chosen based on the type of fibers used for the mesh material **110** and the particular species of ceramic particles used. Infusion conditions such as concentration of ceramic particles in the slurry or solution, solution temperature, slurry temperature, the pH of the solution, the pH of the slurry, contact time of the fibers with the solution or slurry, and amount of current applied to the solution can be chosen based on the type of fibers and species of ceramic particles.

(17) Fabric that is infused with a ceramic mineral powder can offer therapeutic benefits to horses. The types of ceramic mineral powders can create thermally induced photoluminescence emitting light with a wavelength in the “far-infrared” region of the electromagnetic spectrum. Simple body heat is enough thermal energy to induce the infused ceramic powder in the fabric to emit the “far-infrared” light. Infrared light, as well as far-infrared light, has shown the ability to help cells regenerate and repair themselves as well as improve circulation of oxygen-rich blood through the body. Oxygen-rich blood is often needed in areas of the body experiencing pain and deep tissue injuries. By helping improve the flow of oxygen-rich blood, infrared light can promote faster healing and pain relief. Infrared light has also been shown to help reduce inflammation. Infrared light is produced by the sun but is not harmful, in contrast to ultraviolet light. The infrared light is absorbed by photoreceptors in cells, which can then trigger numerous natural processes of the cells offering the aforementioned benefits. Infrared light therapy has been applied to treating back pain, arthritis, bursitis, muscle strain, neck pain, tendonitis, blunt trauma, and more in humans.

(18) Research has shown that infrared light therapy can be just as beneficial for horses as humans. Infrared light used in therapy for humans penetrates about 2 to 7 centimeters under the skin to

reach muscles. While this can work on humans, horses have many areas where deep tissue is deeper than 7 centimeters. Far-infrared light has been shown to penetrate deeper than 7 centimeters making far-infrared light the ideal choice for use in horses. Embodiments of the therapeutic covering **100** can have a far-infrared emissivity wavelength equal to or greater than 8 micrometers. A far-infrared emissivity wavelength in the range of 8 to 14 micrometers has been proven to be most beneficial for health, and embodiments disclosed of the mesh material **110** disclosed herein can have a far-infrared emissivity wavelength in the range of from about 8 to about 14 micrometers. It has been shown that humans, and certain animals including horses, feel far-infrared light simply as heat.

(19) The therapeutic covering **100** made of a mesh material **110** infused with ceramic mineral powder can offer numerous therapeutic benefits to the horses. As mentioned above infrared light and far-infrared light has been shown to help promote cell repair and pain relief in the horses. When the therapeutic covering **100** is placed over the horse, the body heat of the horse can activate the ceramic minerals infused into the mesh material **110**. When activated, the ceramic minerals begin emitting far-infrared light that can gently heat the horse's muscles and promote blood flow throughout the body.

(20) In embodiments, the therapeutic covering **100** can include a plurality of attachment members **160**. Each of the attachment members **160** can be connected to the mesh material **110** and can be adapted to connect spaced portions of the mesh material **110** such that the mesh material **110** can be secured in place upon the horse's body. The plurality of attachment members **160** can be formed as adjustable strap members having releasable connectors **170** attached thereto. The plurality of attachment members **160** can be made of polyester, nylon, acrylic, cotton, or combinations thereof. The plurality of attachment members **160** can be designed to secure two ends of the therapeutic covering **100** together, as seen with the front of the therapeutic covering **100** in FIG. **2**; alternatively, the plurality of attachment members **160** can be designed to encircle the horse and secure the two ends together, but also secure the therapeutic covering **100** to the horse as depicted in FIGS. **1** and **2**. The releasable connectors **170** can be buckles, clasps, ties, hook and loop, or other securing mechanisms. The releasable connectors **170** can be made of plastic, metal, wood, fabric, or combinations thereof.

(21) In addition to being infused with ceramic minerals, the therapeutic covering **100** can also have magnets **120** coupled to the mesh material **110**. The locations of the magnets **120** can correspond to acupuncture points of the horse when the therapeutic covering **100** is placed on the horse, to offer further therapeutic benefits and combination therapy to the horse. The magnets **120** can be coupled to the mesh material **110** by attachment, such as by sewing, glue, hook and loop, clips, or combinations thereof. Alternatively, the magnets **120** can be placed in pockets formed on the therapeutic covering **100**, as is described for FIG. **3** below.

(22) FIG. **3** is an isolated plan view illustrating a pocket **130** of the therapeutic covering **100** having a magnet **120** placed therein. While one pocket **130** is illustrated for one magnet **120** in FIG. **3**, it is contemplated that any number of pocket and magnet pairs can be utilized with the therapeutic covering **100**. The pocket **130** can be of a size just large enough to accept one of the magnets **120**, so that the magnets **120** do not slide or move from the designated acupuncture point when the therapeutic covering **100** is placed over the horse. In embodiments, the fabric of the pocket **130** can be sewn to enclose the magnet **120** into the pocket **130**. The pocket **130** can be made of the same material as the mesh material **110**, or another suitable material. In embodiments of the therapeutic covering **100**, the magnets **120** can be removeable. In additional or alternative embodiments, the therapeutic covering **100** can have multiple sets of magnets **120** of different strengths that may be placed on the therapeutic covering **100** based on the needs of the horse at the time.

(23) In embodiments, each of the magnets **120** can be neodymium magnets. Neodymium magnets are made from an alloy of neodymium, iron, and boron. Neodymium magnets are the strongest type of permanent magnet that is available commercially. Alternatively, other rare-earth magnets can be

used for any of the magnets **120**. Other rare-earth metal magnets may be used to achieve specific strengths or to control a manufacturing cost of the therapeutic covering **100**.
(24) In embodiments, the therapeutic covering **100** can have about 80 to about 150 magnets **120**. Alternatively, more magnets **120** can be used in embodiments of the therapeutic covering **100**. Other embodiments of the therapeutic covering **100** can have fewer magnets **120** to reduce the cost, weight, or provide magnet therapy to the acupuncture points on only specific regions of the horse. The number of magnets **120** used on a given therapeutic covering **100** can depend on the size of the therapeutic covering **100** and the size of the horse or pony for which the therapeutic covering **100** is designed.
(25) In embodiments, one or more of the magnets **120** can have a strength in a range of about 1100 to about 2400 Gauss. In other embodiments, one or more of magnets **120** can have a strength lower than 1100 Gauss for smaller horses and ponies or if the targeted problem is closer to the surface and weaker magnets can suffice. Other embodiments can also have magnets with a strength higher than 2400 Gauss. Magnets with a strength as high as 5000 Gauss can be used safely for therapy, and embodiments of the therapeutic covering **100** can utilize magnets with a strength of up to 5000 Gauss. Magnets used in therapy for humans often are in a range of 300 to 500 Gauss.
(26) The magnets **120** are placed over the primary acupuncture points including, but not limited to: release points that line the vertebrae, ulcer-prone areas and stomach, shoulders, haunches, and rib cages. FIG. **1** shows the plurality of magnets **120** can be placed on acupuncture points including Bladder channel points: BL **13**, BL **14**, BL **17**, BL **22**, BL **26**, BL **27**, BL **29**, BL **54**, BL **36**, BL **37**, BL **38**, BL **39**, BL **42**, BL **46**; Gall Bladder channel points: GB **24**, GB **25**, GB **26**, GB **27**, GB **29**, GB **30**; Small Intestine channel points: SI **13**; SI **10**, SI, **11**, SI **9**; Spleen channel points: SP **21**, SP **13**, SP **20**; Lung channel points: LU **1**; Stomach channel points: ST **32**, ST **34**, ST **35**; Liver channel points: LV **13**, LV **14**; Pericardium channel points: PC **1**; and Triple Heater channel points: TH **10**, TH **13**. The magnets **120** can also be placed on the side of the therapeutic covering **100** not depicted in FIG. **1** over similar acupuncture points as those depicted. The therapeutic covering **100** is cost-effective to produce in the embodiments.
(27) The therapeutic covering **100** uses a combination of the ceramic mineral powder infused in the mesh material **110** and the attached magnets **120** to offer combination therapy to horses in a non-invasive manner. Competition horses are tested for certain substances that violate the competition rules (steroids or similar performance enhancing drugs) during said competitions. Medication designed to aid the competition horses with things such as back-soreness relief and reducing inflammation can be detected in these tests. If detected, the horse can be expelled from the competition. The therapeutic covering **100** offers an alternative to these drugs that cannot be detected in the blood or urine tests by offering relief through combination therapy that does not involve administering substances that enter the horse's bloodstream. Embodiments of the therapeutic covering **100** can have extensions to fit over a horse's tack so that the therapeutic covering **100** can be left on the horse, offering therapy and relief, until the rider mounts the horse. As the therapeutic covering **100** offers breathability and insect resistance, the therapeutic covering **100** is also able to be worn and offer relief to the horse during transport to the competition.
(28) Also disclosed herein is a method for ceramic and magnetic therapy of a horse. The method can include placing the therapeutic covering **100** on a horse; and allowing the therapeutic covering **100** to perform ceramic and magnetic therapy on the horse. The therapeutic covering **100** can be of any embodiment disclosed hereinabove. The method can further include positioning the therapeutic covering **100** on the horse such that the magnets **120** are positioned to correspond with acupuncture points of the horse. The method achieves the unexpected and surprising results disclosed above via use of the therapeutic covering **100** for the ceramic and magnetic therapy of the horse.
(29) The embodiments of the invention described herein are exemplary and numerous modifications, variations and rearrangements can be readily envisioned to achieve substantially equivalent results, all of which are intended to be embraced within the spirit and scope of the

invention. Further, the purpose of the foregoing abstract is to enable the U.S. Patent and Trademark Office and the public generally, and especially the scientist, engineers and practitioners in the art who are not familiar with patent or legal terms or phraseology, to determine quickly from a cursory inspection the nature and essence of the technical disclosure of the application.

Claims

1. A therapeutic covering for a horse, the therapeutic covering comprising: a mesh material infused with a ceramic mineral powder and configured to cover at least a portion of the horse, wherein the mesh material has a mesh size in a range of greater than 2.5 millimeters and less than 5 millimeters; and a plurality of magnets coupled to an outer surface of the mesh material.
2. The therapeutic covering of claim 1, wherein the mesh size is in a range of from 3 millimeters to 4 millimeters.
3. The therapeutic covering of claim 2, wherein the mesh size is 3.5 millimeters.
4. The therapeutic covering of claim 1, wherein the mesh material is a net of knitted strands of fibers.
5. The therapeutic covering of claim 4, wherein the fibers are made of polyester, polypropylene, nylon, micro modal, bamboo, wool, or combinations thereof.
6. The therapeutic covering of claim 4, wherein the net comprises a warp-knitted net.
7. The therapeutic covering of claim 1, wherein each of the plurality of magnets is a neodymium magnet having a strength in a range of 1100 to 2400 Gauss.
8. The therapeutic covering of claim 7, wherein the plurality of magnets comprises from 80 to 150 magnets.
9. The therapeutic covering of claim 1, wherein the plurality of magnets are positioned on the outer surface of the mesh material to correspond with acupuncture points of the horse when the therapeutic covering is placed on the horse.
10. The therapeutic covering of claim 1, further comprising a plurality of pockets formed on the outer surface of the mesh material, wherein each of the plurality of pockets is configured to receive and hold one of the plurality of magnets.
11. The therapeutic covering of claim 1, wherein the ceramic mineral powder is made of silica, alumina, magnesia, or a combination thereof.
12. The therapeutic covering of claim 1, wherein the ceramic mineral powder is configured to emit far-infrared light at a wavelength of greater than 8 micrometers when the therapeutic covering is placed on the horse.
13. The therapeutic covering of claim 1, further comprising: a plurality of attachment members coupled to the mesh material, the attachment members having a corresponding number of releasable connecters.
14. The therapeutic covering of claim 1, wherein the mesh material is a net of knitted strands of fibers.
15. The method of claim 14, wherein the fibers are made of polyester, polypropylene, nylon, micro modal, bamboo, wool, or combinations thereof.
16. The method of claim 14, wherein the ceramic mineral powder is made of silica, alumina, magnesia, or a combination thereof; and wherein each of the plurality of magnets is a neodymium magnet.
17. A method for ceramic and magnetic therapy of a horse, the method comprising the steps of: placing a therapeutic covering on the horse; and allowing the therapeutic covering to perform ceramic therapy and magnetic therapy on the horse; wherein the therapeutic covering comprises i) a mesh material infused with a ceramic mineral powder and configured to cover at least a portion of a horse, wherein the mesh material has a mesh size in a range of greater than 2.5 millimeters and less than 5 millimeters, and ii) a plurality of magnets coupled to an outer surface of the mesh

material.
18. The method of claim 17, wherein the mesh size is in a range of from 3 millimeters to 4 millimeters.
19. The method of claim 18, wherein the mesh size is 3.5 millimeters.
20. The method of claim 17, further comprising: positioning the therapeutic covering on the horse such that the plurality of magnets are positioned to correspond with acupuncture points of the horse.